

DEVAYAN MANDAL

Full Stack AI Native / Forward Deployed AI Engineer & Open Source Contributor - Open to Remote Work

devayanmandal@gmail.com

linkedin.com/in/dev-ma | github.com/devtechedge

## PROFESSIONAL SUMMARY

Forward Deployed AI Engineer with 10 years of self-directed production work and 30+ shipped systems across multi-agent platforms, MCP integrations, deterministic RL evaluation, and real-time infrastructure. Merged 21 upstream pull requests across TypeScript, Python, and Rust in widely used projects including Biome, pnpm, and node-postgres, owning each change from root cause through tests, review, and release.

## TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Rust, SQL, R, Bash

AI & Agents: LangGraph, LangChain, LangServe, LangSmith, Model Context Protocol (MCP), multi-agent orchestration, supervisor routing, Reflexion loops, human-in-the-loop gates, hybrid RAG, RLVR environments, deterministic evaluation, prompt engineering, Ollama, Gemini, OpenAI-compatible providers

Frontend: React 19, Next.js 16, TanStack Start/Router/Query, Tailwind v4, Vite, Framer

Motion, Expo, Three.js, React Three Fiber, HTML5 Canvas, SVG

Backend & Data: FastAPI, Node.js, Drizzle ORM, Prisma, SQLAlchemy, PostgreSQL (pgvector), Neon, Supabase, SQLite, Pydantic, Zod, Stripe

Realtime & Observability: SSE, WebSockets, Socket.io, replayable traces,

OpenTelemetry-shaped tracing, LangSmith evaluation pipelines

Cloud & DevOps: Vercel, AWS, Microsoft Azure, GitHub Actions CI/CD, serverless, scheduled pipelines, Vitest, pytest, Playwright, Biome

Web3 & Blockchain: Solidity, OpenZeppelin, Foundry, viem, wagmi, ethers.js, Safe Protocol

Kit, RainbowKit, Reown AppKit, Base L2, blockchain and wallet infrastructure

Practices: Six Sigma DMAIC, bounded autonomy, retrieval-bounded generation, AST-sandboxed grading, cost-aware routing, threat-model documentation, Git, Changesets

## PROFESSIONAL EXPERIENCE

### LATTICE AI LABS

Full Stack AI Native / Forward Deployed AI Engineer | January 2016 - Present | Remote

- Architected and shipped 30+ production-grade systems spanning autonomous agents, RL evaluation environments, real-time observability, and high-throughput web applications.
- Built AEGIS, an enterprise AI ops cortex where a LangGraph supervisor routes 6 specialist agents across 7 agentic loops, with hybrid RAG, human-in-the-loop gates, LangSmith evaluation gates, and a live SSE dashboard. Live: [aegis-agent-api.vercel.app/ui](https://aegis-agent-api.vercel.app/ui)
- Built Synthesis, an autonomous research platform running cyclic Reflexion workflows with HITL approval gates, and AAROP, an inspectable 6-phase reasoning engine with cost-aware routing, circuit breakers, RAG memory, and replayable traces. Live: [synthesis-gold.vercel.app](https://synthesis-gold.vercel.app) | [aarop.vercel.app](https://aarop.vercel.app)
- Authored 3 official RLVR environments on the Prime Intellect Environments Hub (calendar-math, meeting-slot, unit-fix) using deterministic UTC sweep-line solvers and AST-sandboxed graders, removing LLM-as-judge variance.
- Implemented MCP servers and clients across Stdio and StreamableHTTP/SSE transports with client-side sampling and roots-based security boundaries; shipped HoldSlot, a WebMCP implementation for the OpenAI WebMCP Challenge 2026.
- Shipped Jobrow, a live US tech job board indexing 5,000+ still-open roles across employer ATS boards (Greenhouse, Ashby, Lever) on TanStack Start, React 19, and Neon Postgres, with an Expo mobile client and a GitHub Actions crawl pipeline running 4x daily. Live: [jobrow.vercel.app](https://jobrow.vercel.app)
- Also delivered Veritas (LangGraph research agent), Cadence (healthcare agent with 5-layer clinical memory), a real-time chaos engineering platform with self-healing services and sub-second WebSocket telemetry, Pulsar (Base L2 GPU marketplace on Foundry contracts), and a serverless daily AI news agent on GitHub Actions.
- Work AI-native across the full stack using frontier coding and reasoning models, with emphasis on rapid iteration, test-driven validation, repository-specific contribution policies, and reproducible upstream delivery.

## SPECTRAL QUANTITATIVE TRADING

Quantitative Crypto Trader & Derivatives Strategist | September 2016 - Present | Remote

- Manage diversified proprietary capital across tier-1 exchanges including Binance, Coinbase, Bitget, and MEXC, and on-chain protocols including Hyperliquid, Uniswap v3, Aave, and GMX.
- Built and ran automated grid, DCA, and Martingale bots on Pionex and Binance to capture spread and volatility in ranging markets.
- Executed delta-neutral basis and funding-rate arbitrage between spot and perpetuals, plus concentrated-liquidity and lending-loop strategies on-chain.
- Applied TWAP/VWAP execution on illiquid pairs, with VaR limits, drawdown caps, and Sharpe-ratio optimization governing position sizing and portfolio risk.

## OPEN SOURCE CONTRIBUTIONS

Merged upstream:

<<<LEDGER:MERGED\_LIST>>>

- leo-aa88/reviewgate #144 (Python) - adds a configurable excessive code-comment verbosity check to the core review engine: three signals computed from added patch lines only, so a PR is never penalised for comments it did not touch, with a single-pass lexer that tracks string literals, heredocs and template literals so runtime data is never counted as commentary. Merged Sep 2026. [github.com/leo-aa88/reviewgate/pull/144](https://github.com/leo-aa88/reviewgate/pull/144)
- PyO3/maturin #3302 (Rust) - Windows interpreter discovery no longer panics on non-UTF-8 command output. `find_all_windows()` decodes `py --list-paths` and `conda info -e` with `String::from_utf8_lossy` instead of unwrapping `str::from_utf8`, so a localized install that emits undecodable bytes skips that line instead of aborting the build. Added a unit test covering the non-UTF-8 decode path. Merged Sep 2026. [github.com/PyO3/maturin/pull/3302](https://github.com/PyO3/maturin/pull/3302)
- stellar/stellar-docs #2859 (MDX) - added the `--enable-core-manual-close` flag to the Local section of the Network Modes page, so the parameter list no longer omits a flag the quickstart container ships, with a cross-link to the Manual close mode section noting that only the local configuration sets `NODE_IS_VALIDATOR`. Merged Sep 2026. [github.com/stellar/stellar-docs/pull/2859](https://github.com/stellar/stellar-docs/pull/2859)
- stellar/stellar-docs #2850 (MDX) - documented Quickstart's undocumented `'--enable-core-manual-close'` flag in the advanced usage docs, covering the `'MANUAL_CLOSE'` setting it writes into the generated core config, the `'manualclose'` endpoint that advances one ledger per call, and loopback-bound startup examples for macOS, Linux, and Windows. Merged Sep 2026. [github.com/stellar/stellar-docs/pull/2850](https://github.com/stellar/stellar-docs/pull/2850)
- ssf0409/tracelens #140 (Python) - markdown table cells in the report generator are now escaped with `html.escape` in addition to pipe and newline handling. Task ids or gate values containing pipes, line breaks, or HTML metacharacters no longer break the per-task and baseline-gate tables piped into `$GITHUB_STEP_SUMMARY`, and no cell can open a raw HTML element. Merged Sep 2026. [github.com/ssf0409/tracelens/pull/140](https://github.com/ssf0409/tracelens/pull/140)
- stellar/stellar-docs #2849 (MDX) - reworked the address conversion example to propagate the fallible `'Result<Address, ConversionError>'` instead of unwrapping malformed XDR, and documented that untrusted XDR must be handled rather than panicking. Merged Sep 2026. [github.com/stellar/stellar-docs/pull/2849](https://github.com/stellar/stellar-docs/pull/2849)
- stellar/stellar-docs #2853 (MDX) - reconciled memo framing in the pooled accounts guide so memos read as the legacy mechanism still in active use and muxed accounts as preferred going forward. Merged Sep 2026. [github.com/stellar/stellar-docs/pull/2853](https://github.com/stellar/stellar-docs/pull/2853)
- stellar/stellar-docs #2851 (MDX) - qualified the Freighter HTTPS requirement, noting that `'http://localhost'` is already a Potentially Trustworthy origin under W3C Secure Contexts so loopback needs no TLS. Merged Sep 2026. [github.com/stellar/stellar-docs/pull/2851](https://github.com/stellar/stellar-docs/pull/2851)
- pytest-dev/pytest-env #262 (Python) - documented that `pytest.toml` and `.pytest.toml` accept the native `[pytest]` env table, not only the plugin-specific `[pytest_env]` section. Merged Sep 2026. [github.com/pytest-dev/pytest-env/pull/262](https://github.com/pytest-dev/pytest-env/pull/262)
- pnpm/pnpm #14863 (Rust/TypeScript) - fixed startup crashes on FreeBSD and other non-Windows Unix-like platforms by making `default_store_dir` use the Unix fallback path; added platform-specific regression coverage. Merged Sep 2026. [github.com/pnpm/pnpm/pull/14863](https://github.com/pnpm/pnpm/pull/14863)
- web-infra-dev/rspress #3678 (TypeScript) - fixed a Search initialization race by awaiting `FlexSearch.addAsync` indexing before `LocalProvider.init()` resolves, preventing early searches from incorrectly returning no results on larger sites; focused regression-oriented fix. Merged Sep 2026. [github.com/web-infra-dev/rspress/pull/3678](https://github.com/web-infra-dev/rspress/pull/3678)
- recharts/recharts #7805 (TypeScript) - removed `tabIndex=-1` from z-index portal SVG `<g>` layers so empty groups are not pointer-focusable and WebKit does not draw geometry-traced focus rings; added regression coverage asserting no layer carries a `tabindex` attribute.

Merged Sep 2026. [github.com/recharts/recharts/pull/7805](https://github.com/recharts/recharts/pull/7805)  
- [brianc/node-postgres #3772](https://github.com/brianc/node-postgres/pull/3772) (JavaScript) - stopped `Connection.sync()` from incorrectly setting the internal `_ending` flag, preventing false suppression of subsequent socket errors; added unit coverage for Sync vs disconnect semantics. Merged Sep 2026. [github.com/brianc/node-postgres/pull/3772](https://github.com/brianc/node-postgres/pull/3772)  
- [pnpm/pnpm #14756](https://github.com/pnpm/pnpm/pull/14756) (Rust/TypeScript) - preserved existing dependency range operators and `npm:/jsr:` protocol prefixes during pnpm update, including prerelease and exact-version handling. Merged Sep 2026. [github.com/pnpm/pnpm/pull/14756](https://github.com/pnpm/pnpm/pull/14756)  
- [pnpm/pnpm #14754](https://github.com/pnpm/pnpm/pull/14754) (Rust) - fixed non-recursive pattern runs with `--no-bail` so matching scripts continue executing and failures are aggregated correctly; added concurrent and sequential regression tests. Merged Sep 2026. [github.com/pnpm/pnpm/pull/14754](https://github.com/pnpm/pnpm/pull/14754)  
- [pnpm/pnpm #14753](https://github.com/pnpm/pnpm/pull/14753) (Rust/TypeScript) - fixed `lockfile:false` being ignored when automatic package-manager switching downloaded the configured engine; added unit and filesystem-backed regression coverage across Rust and TypeScript paths. Merged Sep 2026. [github.com/pnpm/pnpm/pull/14753](https://github.com/pnpm/pnpm/pull/14753)  
- [SQLMesh/sqlmesh #6040](https://github.com/SQLMesh/sqlmesh/pull/6040) (Python) - moved `ModelTest.create_test()` off concurrent worker threads to eliminate a `time_machine/ttl_cache` race in sqlmesh test; verified the focused race regression repeatedly. Merged Sep 2026. [github.com/SQLMesh/sqlmesh/pull/6040](https://github.com/SQLMesh/sqlmesh/pull/6040)  
- [anza-xyz/kit #2032](https://github.com/anza-xyz/kit/pull/2032) (TypeScript) - restored number predicates in the `getPatternMatchCodec` advanced guide to match current codec typing. Merged Sep 2026. [github.com/anza-xyz/kit/pull/2032](https://github.com/anza-xyz/kit/pull/2032)  
- [better-auth/better-auth #11208](https://github.com/better-auth/better-auth/pull/11208) (TypeScript) - added regression coverage locking the `/phone-number/verify` OpenAPI `requestBody` after a `Zod.and(z.record())` intersection; tests only, no production generator changes. Merged Sep 2026. [github.com/better-auth/better-auth/pull/11208](https://github.com/better-auth/better-auth/pull/11208)  
- [biomejs/biome #11667](https://github.com/biomejs/biome/pull/11667) (Rust) - ported the unicorn better-dom-traversing rule into Biome's nursery linter; +1,972 lines across 17 files and 7 commits, released to 13 packages via `changeset`. Merged Sep 2026. [github.com/biomejs/biome/pull/11667](https://github.com/biomejs/biome/pull/11667)  
- [thirdweb-dev/js #8938](https://github.com/thirdweb-dev/js/pull/8938) (TypeScript) - fixed `useTokenQuery` error handling so 401/429/timeout failures rethrow instead of rendering a false "Token Not Supported" state in `CheckoutWidget`; added tests and a patch `changeset`. Merged Sep 2026. [github.com/thirdweb-dev/js/pull/8938](https://github.com/thirdweb-dev/js/pull/8938)  
<<<END:LEDGER:MERGED\_LIST>>>

Canonical ledger with links to every merged PR: [github.com/devtechedge/oss-contributions](https://github.com/devtechedge/oss-contributions)

## CERTIFICATIONS

- Model Context Protocol: Advanced Topics - Anthropic, 2026
- AI Fluency for Builders - Anthropic (with CodePath), 2026
- Agents and Workflows - OpenAI Academy, 2026
- Deep Agents with LangGraph - LangChain Academy, 2026
- Ambient Agents with LangGraph - LangChain Academy, 2026
- Microsoft Certified: Azure Fundamentals (AZ-900) - Microsoft, 2020
- AWS Fundamentals Specialization - Amazon Web Services (Coursera), 2020
- Data Science Specialization - Johns Hopkins University (Coursera), 2020
- Six Sigma Green Belt - Kennesaw State University (Coursera), 2020

## EDUCATION

### UNIVERSITY OF MUMBAI

Bachelor of Engineering (B.E.), Chemical Engineering (First Class) | [Graduation Year]  
Quantitative core in process dynamics, transport phenomena, mathematical modeling, and optimization; transitioned into software engineering through computational and systems-oriented work.